

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Medical image analysis often operates in the quantitatively unfavorable regime where robust vision models are expected to learn from limited expert annotation. Self-supervised masked image modeling (MIM) reduces this dependence on labels, but the dominant random, high-ratio masking recipe is not designed to prioritize the small image regions that drive downstream medical metrics. Existing region-aware variants improve this framing only partially, because they usually depend on reconstruction heuristics or supervised signals that are less portable across modalities, tasks, and evaluation settings. We propose Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The method uses vision-language models to localize prompt-specified regions and then applies differentiated masking so that diagnostically relevant regions receive greater emphasis while background redundancy is reduced. This design makes the comparison target explicit: MWM changes the masking policy, not the downstream labels, and aims to improve representation quality under matched pretraining and fine-tuning conditions. Across brain MRI, chest CT, and lung X-ray evaluations, Mask What Matters improves over existing MIM methods, including SparK, with margins reaching +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection and segmentation transfer. The gains are especially practical because they are obtained with substantially lower overall masking ratios, such as 40% for MWM compared with the 70% setting favored by SparK. These results show that controllable, text-driven masking provides a clearer and more efficient empirical path to semantically aligned self-supervised learning for medical vision models.

1 INTRODUCTION

Image classification, segmentation, and object detection are now routinely evaluated with quantitative metrics that reward both representation quality and task transfer. In medical image processing, however, the empirical setting is constrained by annotation cost: high-performing deep models generally require large labeled datasets, while expert labels are scarce, expensive, and unevenly available across modalities. This gap between metric-driven model development and limited annotation supply motivates pretraining strategies that can improve downstream accuracy and localization performance without increasing labeling burden.

Self-supervised learning (SSL) directly addresses this data-efficiency problem by pretraining on unlabeled images and transferring the encoder to labeled downstream tasks [Jaiswal et al. \(2020\)](#); [Liu et al. \(2021\)](#); [Krishnan et al. \(2022\)](#); [Huang et al. \(2023\)](#). Within SSL, MIM is attractive because a simple reconstruction objective can produce features that improve classification, segmentation, and detection metrics after fine-tuning. Standard MIM randomly masks image patches and reconstructs missing content from visible context, a recipe that has performed well in natural images and has also been adapted to medical image analysis [He et al. \(2022\)](#); [Gupta et al. \(2025\)](#); [Xiao et al. \(2023\)](#).

The main quantitative weakness of prevailing MIM methods, including masked autoencoders (MAE) [He et al. \(2022\)](#) and SparK [Tian et al. \(2023\)](#), is that their random masking policy is not

matched to the regions that determine downstream scores. This task-agnostic design creates two comparison-relevant limitations: (1) representations can be learned from regions that are weakly related to the downstream objective, reducing transfer performance under classification or localization metrics. (2) To raise the probability that informative regions are masked, these methods often use very high masking ratios, such as 75% in MAE, which increases reconstruction difficulty and resource demand. By contrast, natural language has compact structure and high semantic density, enabling effective learning even at low masking ratios [Devlin et al., 2019]. The high-ratio convention in image MIM therefore reflects both visual redundancy and the absence of a semantic mechanism for selecting the regions that matter most for the measured task.

The mismatch is amplified in medical imaging because the signal that drives a correct prediction or localization score is often spatially sparse. Lesions, organs, or abnormal tissue may occupy a small fraction of an MRI, CT, or X-ray image, while much of the field of view contributes redundant background information. The relevant region also changes substantially across modalities and tasks, so a useful masking policy must deliver consistent empirical gains while adapting to different shapes, sizes, and spatial distributions of task-relevant anatomy.

We address this comparison gap with a controllable, text-guided masking framework, **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary description or phrase, a pretrained vision-language model provides zero-shot localization cues, which are converted into robust regions of interest (ROIs). MWM then applies prompt-conditioned, region-specific masking by assigning higher ratios to semantically important areas, such as lesions or organs, and lower ratios to background. The result is a pretraining signal that makes downstream semantics explicit without using per-image reports or labels.

Our contributions are organized around the empirical comparison that this design enables:

- We introduce the first controllable text-guided masking framework for medical imaging. MWM integrates vision-language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and compare differentiated ROI vs. background masking against random masking.
- We provide an annotation-free and backbone-agnostic design. MWM requires no per-image reports or labels, and the same protocol can be used with both ViT- and ConvNet-based MIM pipelines, supporting controlled comparisons across model families.
- We report consistent gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM surpasses state-of-the-art MIM baselines, including SparK, improving classification, detection, and segmentation metrics while using a more semantically targeted masking policy.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling first produced strong quantitative evidence in natural language processing through bidirectional encoder representations from transformers (BERT) [Devlin et al., 2019], which learns contextual representations by predicting masked tokens during pretraining. [He et al., 2022] transfer this principle to vision by masking image regions and optimizing reconstruction, after which the encoder can be compared on downstream classification, segmentation, and detection tasks. This paradigm has shown strong transfer in both natural and medical image domains [Gupta et al., 2025, Liu et al., 2021, Huang et al., 2023]. Representative baselines include MAE [He et al., 2022], MedMAE [Gupta et al., 2025, Xiao et al., 2023], and SparK [Tian et al., 2023]: MAE uses high-ratio random masking with ViTs, MedMAE adapts the same evidence base to medical images, and SparK extends MIM to convolutional backbones by exploiting multi-scale features.

Despite these advances, conventional MIM comparisons are still dominated by random or structure-agnostic masking policies. Because these policies do not distinguish the regions that determine downstream metrics from redundant background, their transfer gains can be diluted by reconstruction effort spent on less relevant content. Recent region-aware masking work therefore asks a more targeted empirical question: whether selecting regions before masking improves semantic alignment and downstream performance. Two main directions have emerged:

One line of work uses loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al. (2023a)] and AnatoMask [Li et al. (2024)] estimate reconstruction difficulty across regions and assign higher masking probabilities to patches with larger losses. This can improve attention to difficult structures, but reconstruction loss is only an indirect proxy for downstream value: in brain magnetic resonance imaging (MRI), complex structures such as the spinal cord or eyeballs may receive high masking priority even when tumor detection metrics depend on different regions.

A second line of work adds external perceptual modules. FocusMAE [Basu et al. (2024)], for example, uses pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. This makes the comparison more semantically focused, but the requirement for supervised, domain-specific RPN training limits how cleanly the approach can be transferred across datasets and tasks without additional annotation.

Together, these results motivate a masking mechanism whose selection criterion is closer to the downstream semantic target while avoiding extra labels. Natural language is well suited for this role because it can specify the comparison-relevant anatomy or abnormality in an interpretable, open-vocabulary form, enabling task-aware masking without per-image supervision.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid improvement of vision-language alignment models has made zero-shot, open-vocabulary region identification increasingly reliable [Radford et al. (2021)]. CLIP-style models learn a shared embedding space for images and text from large-scale image-text pairs, making it possible to compare visual regions against textual concepts without task-specific labels. In medical imaging, BiomedCLIP [Zhang et al. (2023)] strengthens this cross-modal alignment for biomedical content and has shown robust performance across downstream applications. Most uses of vision-language pretrained models, however, still report benefits at the supervised downstream stage, such as object detection or semantic segmentation [Zhong et al. (2022); Aleem et al. (2024); Koleilat et al. (2024)]. Much less evidence has been reported for using their open-vocabulary localization ability inside the MIM pretraining loop, where it could directly change which regions are masked and therefore which representations are learned.

In summary, existing region-aware MIM methods improve over purely random masking, but their evidence is constrained by additional supervision, reconstruction-loss proxies, or limited flexibility. At the same time, vision-language models already provide strong downstream evidence for open-vocabulary semantic alignment, yet their ability to guide the MIM pretraining comparison remains underexplored. This creates a direct opportunity to connect textual semantics with low-level masking decisions and to evaluate whether that connection yields more controllable and empirically stronger self-supervised representations.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is designed to isolate the empirical effect of the masking policy. The framework has two core stages: (1) text-guided region localization, which identifies task-relevant regions from an open-vocabulary prompt using a frozen vision-language model, and (2) region-aware masked image modeling, which applies different masking ratios to the localized ROI and the background before reconstruction.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions with a multi-stage pipeline whose output is a spatial ROI used for the masking comparison. It begins with LLM-based prompt generation and uses BiomedCLIP to extract aligned image-text embeddings. A saliency map is produced by the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined with the Segment Anything Model (SAM). The refined mask is then converted into a bounding box with a controllable expansion margin, yielding a robust spatial guide for the subsequent MIM stage.

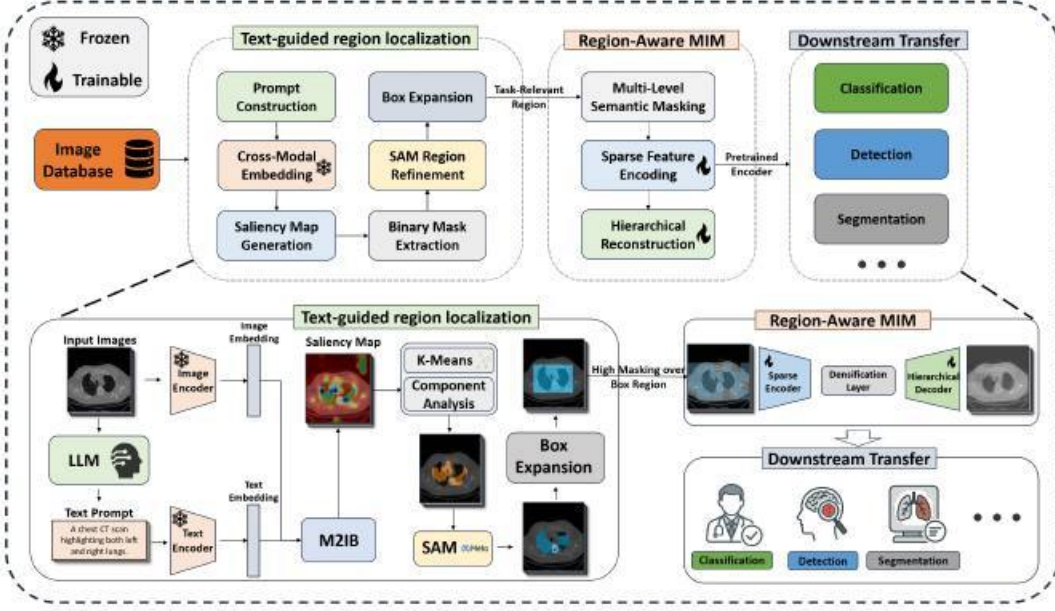


Figure 1: Overview of the MWM comparison pipeline. The top panel separates the three evaluated stages: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel highlights the mechanism used to replace random masking with prompt-guided localization and differentiated ROI/background masking.

Prompt Generation. Prompt design determines which semantic target is compared against the image during localization. To provide explicit medical guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image”. Simple category phrases are also used as alternative inputs, allowing the experiments to quantify how prompt granularity affects localization and downstream accuracy.

Cross-Modal Embedding. After prompt generation, BiomedCLIP [Zhang et al., (2023)] encodes the medical image and prompt in a shared semantic space so that visual evidence can be scored against the same textual target. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These embeddings define the cross-modal comparison used to estimate which image regions are most relevant to the prompt.

Saliency Map Generation. A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al., (2023b)] is used to score visual regions by prompt relevance while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location, making the localization evidence explicit before any masking ratio is applied. M2IB achieves this by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are image and text embeddings encoded by BiomedCLIP respectively, and γ controls the trade-off between retaining cross-modal relevance and filtering visual information that is unlikely to affect the target comparison.

Binary Mask Extraction. After obtaining the saliency map, we binarize it with unsupervised K-Means clustering [Lloyd (1982)] to obtain a preliminary foreground mask for the prompt-guided region. The map is split into two pixel groups, and the group with higher saliency values is treated as foreground. Connected component analysis then retains the largest foreground regions, reducing noisy activations and concentrating the subsequent masking comparison on the core semantic structures.

Region Refinement. To improve the spatial precision of the ROI used for masking, SAM is incorporated as a refinement module [Kirillov et al. \(2023\)](#). For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed to serve as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a more faithful spatial structure for separating ROI and background patches.

Box-Based Region Expansion. The SAM mask obtained from the previous stage may contain boundary noise that would make ROI/background assignment unstable. To keep the masking condition robust and comparable across images, the refined SAM mask is converted into a bounding box with a controllable expansion margin, which serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest generated by the localization module in Section 3.2, MWM performs masked image modeling through a three-stage pipeline that keeps the downstream comparison focused on masking design: (1) multi-level masking applies differentiated ratios according to region importance, (2) a sparse encoder processes only the unmasked patches, and (3) a hierarchical decoder reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM adopts a multi-level masking strategy that uses the semantic regions identified by text-driven localization as the basis for ROI/background comparison. Patches highlighted by textual prompts, such as regions associated with tumors or organs, are masked at a high ratio so that reconstruction emphasizes task-relevant semantic features. Background regions are masked lightly or left unmasked, reducing redundant computation while preserving context. Figure 2 visually contrasts this targeted pattern with conventional masking.

Sparse Feature Encoding. Visible patches are collected into a sparse image and encoded following the SparK design [Tian et al. \(2023\)](#), keeping the encoder-side comparison aligned with a strong MIM baseline:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. Because the sparse interface is unchanged, MWM can be instantiated with convolutional backbones and can also apply to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down manner, keeping the reconstruction objective matched to the masked-patch evidence used by the baselines. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

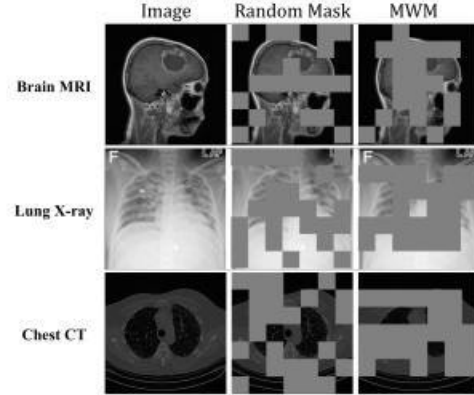


Figure 2: Comparison of masking strategies under the same image context. Gray blocks indicate masked regions, highlighting how MWM concentrates masking on prompt-relevant regions rather than distributing masks randomly.

Finally, the model is trained with mean-squared error on masked patches, so the quantitative difference from prior MIM methods comes from which patches are selected rather than from a new loss:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining uses three datasets selected to test whether the masking policy remains effective across distinct medical image distributions. (1) *Brain MRI* contributes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], providing tumor and anatomical variation for pretraining. (2) *Lung X-ray* provides around 39,000 COVID-QU-Ex chest radiographs [Rahman et al. (2021); Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* supplies 12,000 lung-disease slices [Konya (2020)], together with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation uses three labeled datasets so that representation quality can be compared under classification and detection-oriented transfer settings. (1) *Brain Tumor MRI* [Feltrin (2023)] evaluates 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermay et al. (2018)] evaluates 5,863 pediatric chest X-rays under a normal-versus-pneumonia classification setting. (3) *Chest CT Cancer* [Hany (2020)] evaluates CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are compared at similar model scale: convolution-based methods use ResNet-50 [He et al. (2016)], and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224, models are initialized with ImageNet-pretrained weights, and pre-training uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning keeps the same encoder and input resolution, ensuring that reported accuracy and AP differences primarily reflect the pretraining strategy. All experiments are performed on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables such as prompt type or SAM usage are fixed to their empirically optimal settings for controlled comparison.

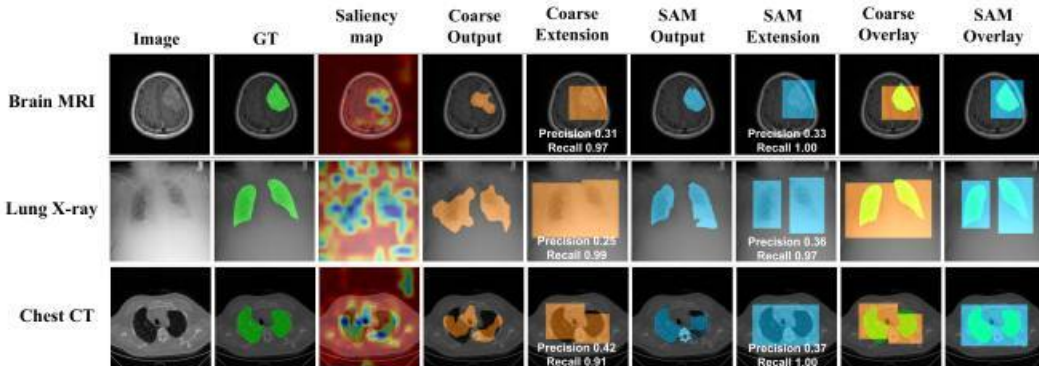


Figure 3: Representative text-guided localization results, illustrating how prompt-derived regions align with the image areas used for ROI-focused masking.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance measured by Occlusion Precision/Recall across prompt types and refinement strategies. The table separates prompt granularity from SAM refinement so the precision–recall trade-off is explicit; *Coarse Ext.* denotes bounding box extension of the coarse region, and + *SAM Ext.* indicates further refinement using SAM.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first quantify whether the text-guided localization module identifies the regions that should receive differentiated masking. Across brain MRI, lung X-ray, and chest CT, predicted bounding boxes are compared with expert-annotated segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation. This evaluation separates localization quality from downstream representation learning.

Table 1 shows that effective text prompts recover task-relevant regions with high recall across modalities while preserving useful precision. The best recall values reach 0.94 for brain MRI, 0.99 for chest CT, and 0.82 for lung X-ray, indicating that textual descriptions can cover the target anatomy despite modality-specific variation. Figure 3 provides representative qualitative examples for these quantitative patterns.

Effect of Prompt Type on Region Localization. To quantify the impact of language design, we compare short phrases with descriptive sentences. In Table 1 descriptive sentences improve precision on all three modalities and improve recall on brain MRI, while chest CT and lung X-ray show higher recall for phrases. For lung radiographs, the recall margin favors phrases by 0.16 (0.82 vs. 0.66), suggesting that broad anatomical terms can be sufficient when the target region occupies a large field of view.

Effect of SAM Refinement on Region Localization. We also isolate the contribution of spatial refinement by comparing coarse box extension with SAM-refined extension. Table 1 shows that SAM improves precision on every modality: from 0.22 to 0.30 on brain MRI, from 0.36 to 0.37 on chest CT, and from 0.37 to 0.42 on lung X-ray. Recall remains high after refinement, including 0.94 on brain MRI and 0.88 on chest CT, supporting SAM as a useful step for tightening ROI definitions before masking.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate whether the localization and masking changes translate into stronger representations. MWM is compared with representative SSL methods from both reconstruction-based and contrastive learning paradigms, with consistent datasets and training settings so that classification accuracy, linear probing accuracy, and AP margins can be interpreted as controlled pretraining comparisons.

Image Classification. Classification performance is examined on three downstream medical imaging datasets under two complementary evaluation settings: (1) *full fine-tuning*, where the entire model is updated and final task accuracy is measured; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained, making representation quality the primary comparison.

- **Full fine-tuning:** Table 2 reports classification accuracy for MWM against representative self-supervised methods, including MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al.

(2020). MWM achieves the highest accuracy on all three datasets, giving a consistent rank advantage across diverse imaging modalities. The largest margin is on Chest CT, where MWM reaches 97.5% accuracy, exceeding SparK by +3.1 points and AnatoMask by +1.7 points. The same pattern also holds on brain MRI and lung X-ray, supporting the practical value of text-guided masking for semantically critical but spatially sparse regions while avoiding redundant masking.

- **Linear probing:** Linear probing further tests whether the encoder itself improves before end-to-end adaptation. Compared with SparK and AnatoMask in Table 3, MWM obtains the best frozen-encoder accuracy on every dataset: 71.3% on brain MRI, 67.0% on chest CT, and 88.5% on lung X-ray. The consistent margins indicate that text-driven pretraining improves feature quality, not only fine-tuning behavior.

Object Detection and Instance Segmentation. To assess transfer beyond classification accuracy, we evaluate brain tumor detection and instance segmentation on the BR35H dataset (Hamada (2025)). As shown in Table 4, MWM achieves the highest AP^{box} at 46.9% and the second-highest AP^{mask} at 45.9%. Relative to SparK, these correspond to +1.3 points in box AP and +1.1 points in mask AP, while AP_{75}^{mask} matches the best value of 54.1%. These results show that the representation gains transfer to both object-level localization and pixel-wise segmentation metrics.

In summary, the empirical pattern is consistent across evaluation forms: text-driven masking improves full fine-tuning accuracy, frozen-encoder linear probing accuracy, and localization-oriented AP metrics. This consistency supports the conclusion that semantic guidance improves the learned representation rather than merely optimizing one downstream protocol.

4.5 ABLATION STUDY

We conduct ablation experiments to quantify which components account for MWM’s downstream classification gains. The ablations separately compare prompt design, SAM-based region refinement, and masking ratio so that each empirical pattern can be interpreted against the random-masking or SparK reference setting.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts achieve the best accuracy on brain MRI and chest CT, reaching 96.8% and 97.5%, while phrases achieve the best lung X-ray accuracy at 96.0%. This modality-specific pattern is consistent with the localization results in Section 4.3: prompt forms that better identify the relevant region also produce stronger downstream classification. Both prompt types outperform the no-prompt random-masking setting on all datasets. The margins are especially clear on Chest CT, where phrase and sentence prompts improve over no prompt by +2.1 and +3.1 points, respectively. The SAM ablation also favors refinement across all modalities, with w/ SAM improving over w/o SAM by +0.4 on brain MRI, +0.7 on chest CT, and +0.2 on lung X-ray, indicating that tighter region definitions translate into measurable accuracy gains.

4.5.2 EFFECT OF MASKING RATIO

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Full fine-tuning classification accuracy (%) on downstream datasets. MIM denotes Masked Image Modeling and CL denotes Contrastive Learning; the table compares MWM against both paradigm families under matched downstream evaluation.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with the encoder frozen, providing a direct comparison of representation quality before full-model adaptation.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection and instance segmentation results (%) on the BR35H dataset. AP^{box} and AP_{75}^{box} measure object detection, while AP^{mask} and AP_{75}^{mask} measure instance segmentation, enabling transfer comparisons beyond classification.

As shown in Figure 4, MWM reaches 96.8% classification accuracy on Chest CT with only 40% masking, whereas SparK’s best evaluated setting occurs at 70% masking. Even against this best SparK configuration, MWM retains a +2.4-point absolute accuracy advantage while using a substantially lower masking ratio. To ensure that the comparison is not driven by an unfavorable SparK setting, SparK is evaluated at 40%, 50%, 60%, and 70% masking ratios. The highest SparK accuracy occurs at 70%, confirming that the reported margin compares MWM against SparK’s strongest ratio among the tested settings.

These results reframe the role of masking ratio in MIM: higher masking is not inherently better when the selected patches are semantically unfocused. MWM achieves stronger downstream accuracy with lower masking because the masked evidence is concentrated on task-relevant content while redundant background is preserved or lightly masked.

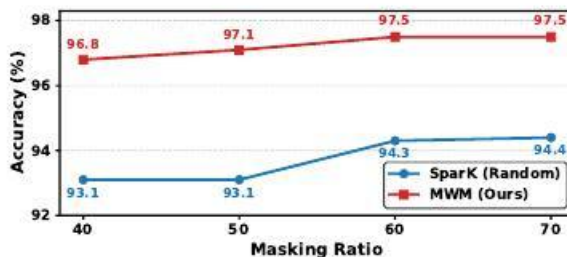


Figure 4: Chest CT classification accuracy under different masking strategies and ratios, showing the practical margin between MWM at 40% masking and SparK across its evaluated masking-ratio range.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. The central empirical finding is that changing which patches are masked, while preserving the underlying SSL setting, yields clearer and stronger downstream metrics than random high-ratio masking. By using natural language prompts to highlight task-relevant areas and applying differentiated masking, MWM provides semantically aligned representation learning without per-image annotations.

Comprehensive experiments support this conclusion across metrics and evaluation conditions: (1) natural language prompts produce measurable localization quality, with recall values reaching 0.94 on brain MRI, 0.99 on chest CT, and 0.82 on lung X-ray; (2) MWM achieves the best classification accuracy across three modalities and improves over strong baselines such as SparK and AnatoMask, while also improving detection and instance segmentation transfer; (3) MWM preserves a practical advantage at lower masking ratios, including 96.8% Chest CT accuracy at 40% masking and a +2.4-point margin over SparK’s best 70% masking result.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under prompt-type and region-refinement ablations across three imaging modalities. *No Prompt* is the random-masking comparison without text-guided localization; *w/* and *w/o* denote with and without SAM, respectively.

Limitations. Although the reported gains are consistent, several quantitative questions remain open. First, the experiments compare phrase and sentence prompts, but they do not systematically measure robustness to broader prompt-style, vocabulary, or noise variations. Second, MWM depends on external textual descriptions; in settings where reliable prompts are unavailable or inconsistent, the localization quality and downstream margins may weaken.

Looking forward, the empirical pattern suggests that text-driven, region-aware masking is a practical route toward more semantically grounded self-supervised learning. Extending MWM beyond medical imaging would test whether the same comparison structure—lower redundant masking, clearer semantic targeting, and consistent downstream margins—also benefits broader vision–language pre-training.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* to edit language and polish presentation, including grammar, phrasing, and minor stylistic clarity, for text written by the authors. LLMs were *not* used to generate ideas, design experiments, annotate data, implement code, or analyze and interpret quantitative results. All technical content, algorithms, proofs, figures, tables, reported metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.